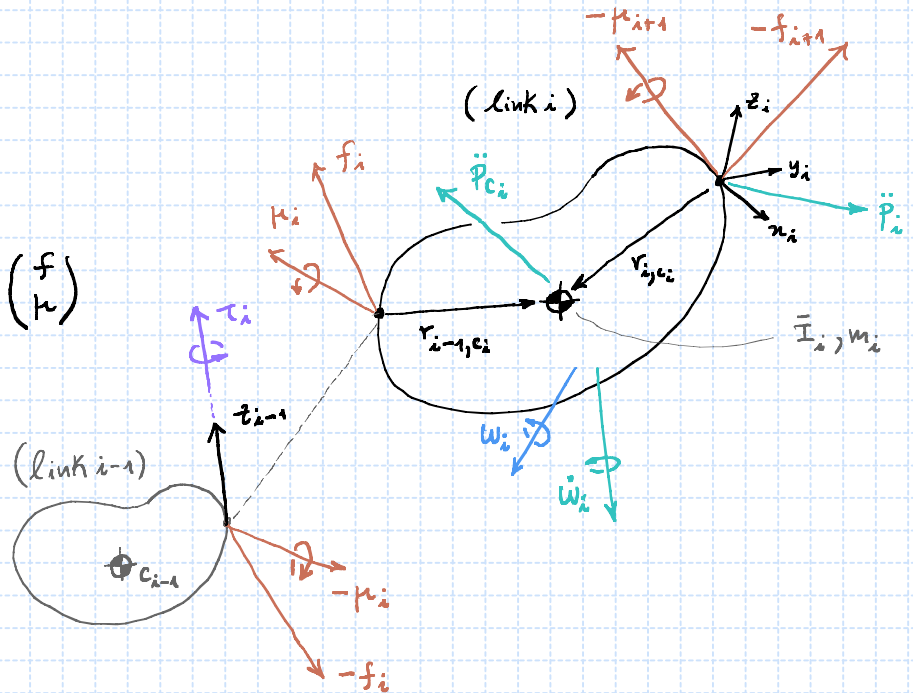


## Newton-Euler

$$\begin{cases} \dot{P} = \Sigma f \\ \dot{H} = \Sigma \mu \end{cases}$$

$$\begin{pmatrix} m & 0 \\ 0 & I_c \end{pmatrix} \begin{pmatrix} \ddot{P}_c \\ \dot{\omega} \end{pmatrix} + \begin{pmatrix} 0 \\ \omega \times (I_c \omega) \end{pmatrix} = \Sigma \begin{pmatrix} f \\ \mu \end{pmatrix}$$



- Forward Recursion  $(\dot{W}_i^i, \dot{\omega}_i^i, \ddot{P}_{c_i}^i)$ ,  $i = 0, 1, 2, \dots, n$

$$\dot{W}_i^i = \begin{cases} R_{i-1}^i \dot{W}_{i-1}^{i-1} & , q_i \text{ is prismatic} \\ R_{i-1}^i (\dot{W}_{i-1}^{i-1} + \dot{\theta}_i \hat{K}) & , q_i \text{ is revolute} \end{cases}$$

$$\dot{\omega}_i^i = \begin{cases} R_{i-1}^i \dot{\omega}_{i-1}^{i-1} & , q_i \text{ is prismatic} \\ R_{i-1}^i (\dot{\omega}_{i-1}^{i-1} + \ddot{\theta}_i \hat{K} + \dot{\theta}_i S(\dot{W}_{i-1}^{i-1}) \hat{K}) & , q_i \text{ is revolute} \end{cases}$$

$$\ddot{P}_i^i = \begin{cases} R_{i-1}^i (\ddot{P}_{i-1}^{i-1} + \ddot{d}_i \hat{K}) + 2\dot{d}_i S(\dot{W}_i^i) R_{i-1}^i \hat{K} + S(\dot{\omega}_i^i) r_{i-1,i}^i + S(\dot{W}_i^i) S(\dot{W}_i^i) r_{i-1,i}^i & , q_i \text{ is prismatic} \\ R_{i-1}^i \ddot{P}_{i-1}^{i-1} + S(\dot{\omega}_i^i) r_{i-1,i}^i + S(\dot{W}_i^i) S(\dot{W}_i^i) r_{i-1,i}^i & , q_i \text{ is revolute} \end{cases}$$

$$\ddot{P}_{c_i}^i = \ddot{P}_i^i + S(\dot{\omega}_i^i) r_{i,c_i}^i + S(\dot{W}_i^i) S(\dot{W}_i^i) r_{i,c_i}^i$$

- Backward Recursion  $(f_i^i, \mu_i^i)$ ,  $i = n, n-1, \dots, 1$

$$f_i^i = m_i \ddot{P}_{c_i}^i + R_{i+1}^i f_{i+1}^{i+1}$$

$$\mu_i^i = \bar{I}_i^i \dot{\omega}_i^i + S(\dot{W}_i^i) \bar{I}_i^i \dot{W}_i^i + R_{i+1}^i \mu_{i+1}^{i+1} + S(r_{i-1,i}^i) f_i^i - S(r_{i,c_i}^i) R_{i+1}^i f_{i+1}^{i+1}$$

- Joint axis projection

$$\tau_i = \begin{cases} f_i^{i,T} R_{i-1}^i \hat{K} & , q_i \text{ is prismatic} \\ \mu_i^{i,T} R_{i-1}^i \hat{K} & , q_i \text{ is revolute} \end{cases}$$

$$S(a) = \begin{pmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{pmatrix}$$

$$a \times b = S(a)b$$

$$S(b)a = -a \times b$$

$$S(a)^T = -S(a)$$

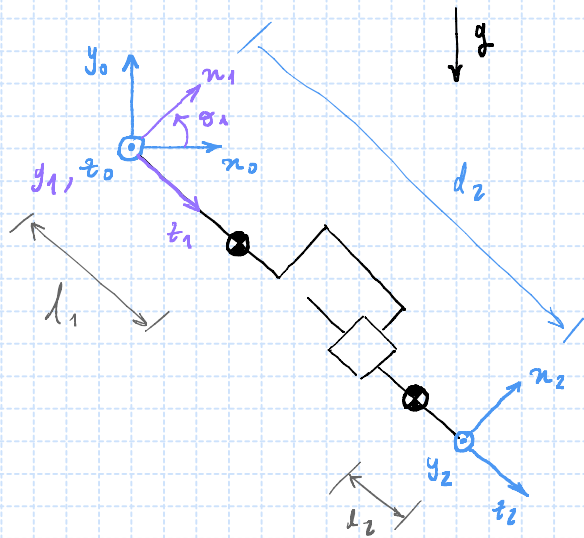
$$RS(a)R^T = S(Ra)$$

$$r_{i-1,i}^i = R_i^T (P_i - P_{i-1})$$

$$P_{c_i} = P_i + R_i r_{i,c_i}^i$$

$$r_{i-1,c_i}^i = R_i^T (P_{c_i} - P_{i-1})$$

## (a) Direct kinematics



$$R_1^0 = \begin{pmatrix} c_1 & 0 & s_1 \\ s_1 & 0 & -c_1 \\ 0 & 1 & 0 \end{pmatrix}, \quad R_2^1 = I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$R_0^1 = \begin{pmatrix} c_1 & s_1 & 0 \\ 0 & 0 & 1 \\ s_1 & -c_1 & 0 \end{pmatrix}, \quad R_1^2 = I.$$

$$r_{i-1,i}^i: r_{0,1}^1 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad r_{1,2}^2 = \begin{pmatrix} 0 \\ 0 \\ d_2 \end{pmatrix}.$$

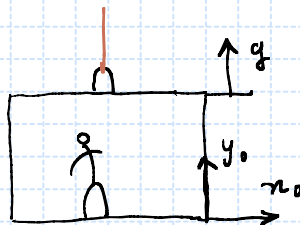
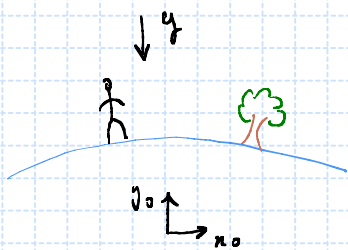
$$r_{i,j,c_i}^i: r_{1,c_1}^1 = \begin{pmatrix} 0 \\ 0 \\ l_1 \end{pmatrix}, \quad r_{2,c_2}^2 = \begin{pmatrix} 0 \\ 0 \\ -l_2 \end{pmatrix}.$$

$$r_{i-1,c_i}^i: r_{0,c_1}^1 = \begin{pmatrix} 0 \\ 0 \\ l_1 \end{pmatrix}, \quad r_{1,c_2}^2 = \begin{pmatrix} 0 \\ 0 \\ d_2 - l_2 \end{pmatrix}.$$

## (b) Forward Recursion (differential kinematics)

$$i=0,$$

$$\omega_0^0 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad \dot{\omega}_0^0 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad \ddot{p}_0^0 = \begin{pmatrix} 0 \\ 0 \\ 1g \end{pmatrix} \approx \begin{pmatrix} 0 \\ 9.81 \\ 0 \end{pmatrix} [\text{m/s}^2].$$



uniformly accelerated  
chest mount - experiment.  
Einstein, 1907.

$$i=1,$$

$$\omega_1^1 = R_0^1 (\omega_0^0 + \dot{\theta}_1 \hat{k}) = \dot{\theta}_1 R_0^1 \hat{k} = \dot{\theta}_1 z_0^1 = \begin{pmatrix} 0 \\ \dot{\theta}_1 \\ 0 \end{pmatrix}.$$

$$\dot{\omega}_i^i = R_{i-1}^i (\dot{\omega}_{i-1}^{i-1} + \dot{\theta}_i \hat{k})$$

$$R_0^1 = \begin{pmatrix} c_1 & s_1 & 0 \\ 0 & 0 & 1 \\ s_1 & -c_1 & 0 \end{pmatrix}$$

$$\ddot{\omega}_i^i = R_{i-1}^i (\ddot{\omega}_{i-1}^{i-1} + \ddot{\theta}_i \hat{k} + \dot{\theta}_i S(\dot{\omega}_{i-1}^{i-1}) \hat{k})$$

$$\ddot{\omega}_1^1 = R_0^1 (\ddot{\omega}_0^0 + \ddot{\theta}_1 \hat{k} + \dot{\theta}_1 \underbrace{S(\dot{\omega}_0^0) \hat{k}}_{\omega_0^0 \times \hat{k}}) = \ddot{\theta}_1 R_0^1 \hat{k} = \begin{pmatrix} 0 \\ \ddot{\theta}_1 \\ 0 \end{pmatrix}.$$

$$\ddot{p}_i^i = R_{i-1}^i \ddot{p}_{i-1}^{i-1} + S(\dot{\omega}_i^i) r_{i-1,i}^i + S(\dot{\omega}_i^i) S(\dot{\omega}_i^i) r_{i-1,i}^i$$

$$\ddot{p}_1^1 = R_0^1 \ddot{p}_0^0 + S(\dot{\omega}_1^1) r_{0,1}^1 + S(\dot{\omega}_1^1) S(\dot{\omega}_1^1) r_{0,1}^1 = \begin{pmatrix} c_1 & s_1 & 0 \\ 0 & 0 & 1 \\ s_1 & -c_1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1g \\ 0 \end{pmatrix} = \begin{pmatrix} 1g s_1 \\ 0 \\ -1g c_1 \end{pmatrix}.$$

$$\ddot{p}_{c_i}^i = \ddot{p}_i^i + S(\dot{\omega}_i^i) r_{i,j,c_i}^i + S(\omega_i^i) S(\omega_i^i) r_{i,j,c_i}^i$$

$$\begin{aligned} \ddot{p}_{c_1}^1 &= \ddot{p}_1^1 + S(\dot{\omega}_1^1) r_{1,1,c_1}^1 + S(\omega_1^1) S(\omega_1^1) r_{1,1,c_1}^1 \\ &= \ddot{p}_1^1 + \underbrace{\begin{pmatrix} 0 & 0 & \ddot{\theta}_1 \\ 0 & 0 & 0 \\ -\ddot{\theta}_1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ l_1 \end{pmatrix}}_{\begin{pmatrix} l_1 \ddot{\theta}_1 \\ 0 \\ 0 \end{pmatrix}} + \underbrace{S(\omega_1^1) \begin{pmatrix} 0 & 0 & \dot{\theta}_1 \\ 0 & 0 & 0 \\ -\dot{\theta}_1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ l_1 \end{pmatrix}}_{\begin{pmatrix} l_1 \dot{\theta}_1 \\ 0 \\ -l_1 \dot{\theta}_1^2 \end{pmatrix}} \\ &= \begin{pmatrix} |g|s_1 + l_1 \ddot{\theta}_1 \\ 0 \\ -|g|c_1 - l_1 \dot{\theta}_1^2 \end{pmatrix} \end{aligned}$$

$$S(a) = \begin{pmatrix} 0 & -a_z & a_y \\ a_z & 0 & -a_x \\ -a_y & a_x & 0 \end{pmatrix}$$

$$a \times b = S(a)b$$

$$\dot{n} = 2,$$

$$\omega_2^2 = \underbrace{R_1^2}_{\mathbb{I}} \omega_1^1 = \omega_1^1 = \begin{pmatrix} 0 \\ \dot{\theta}_1 \\ 0 \end{pmatrix}$$

$$\omega_i^i = R_{i-1}^i \omega_{i-1}^{i-1}$$

$$\dot{\omega}_i^i = R_{i-1}^i \dot{\omega}_{i-1}^{i-1}$$

$$\dot{\omega}_2^2 = R_1^2 \dot{\omega}_1^1 = \dot{\omega}_1^1 = \begin{pmatrix} 0 \\ \dot{\theta}_1 \\ 0 \end{pmatrix}$$

$$\ddot{p}_i^i = R_{i-1}^i (\ddot{p}_{i-1}^{i-1} + \ddot{d}_i \hat{k}) + 2\dot{d}_i S(\omega_i^i) R_{i-1}^i \hat{k} + S(\dot{\omega}_i^i) r_{i-1,i}^i + S(\omega_i^i) S(\omega_i^i) r_{i-1,i}^i$$

$$\begin{aligned} \ddot{p}_2^2 &= \underbrace{R_1^2}_{\mathbb{I}} (\ddot{p}_1^1 + \ddot{d}_2 \hat{k}) + 2\dot{d}_2 \underbrace{S(\omega_2^2) R_1^2}_{\mathbb{I}} \hat{k} + S(\dot{\omega}_2^2) r_{1,2}^2 + S(\omega_2^2) S(\omega_2^2) r_{1,2}^2 \\ &= \ddot{p}_1^1 + \ddot{d}_2 \hat{k} + 2\dot{d}_2 \underbrace{\begin{pmatrix} 0 & 0 & \ddot{\theta}_1 \\ 0 & 0 & 0 \\ -\ddot{\theta}_1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}}_{\begin{pmatrix} \ddot{\theta}_1 \\ 0 \\ 0 \end{pmatrix}} + \underbrace{\begin{pmatrix} 0 & 0 & \ddot{\theta}_1 \\ 0 & 0 & 0 \\ -\ddot{\theta}_1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ d_2 \end{pmatrix}}_{\begin{pmatrix} d_2 \ddot{\theta}_1 \\ 0 \\ 0 \end{pmatrix}} + \underbrace{S(\omega_2^2) \begin{pmatrix} 0 & 0 & \dot{\theta}_1 \\ 0 & 0 & 0 \\ -\dot{\theta}_1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ d_2 \end{pmatrix}}_{\begin{pmatrix} d_2 \dot{\theta}_1 \\ 0 \\ -d_2 \dot{\theta}_1^2 \end{pmatrix}} \\ \ddot{p}_2^2 &= \begin{pmatrix} |g|s_1 + 2\dot{d}_2 \ddot{\theta}_1 + d_2 \ddot{\theta}_1 \\ 0 \\ -|g|c_1 + \ddot{d}_2 - d_2 \dot{\theta}_1^2 \end{pmatrix} \end{aligned}$$

$$\ddot{p}_1^1 = \begin{pmatrix} |g|s_1 \\ 0 \\ -|g|c_1 \end{pmatrix}$$

$$\begin{aligned} \ddot{p}_{c_2}^2 &= \ddot{p}_2^2 + S(\dot{\omega}_2^2) r_{2,1,c_2}^2 + S(\omega_2^2) S(\omega_2^2) r_{2,1,c_2}^2 = \\ &= \ddot{p}_2^2 + \underbrace{\begin{pmatrix} 0 & 0 & \ddot{\theta}_1 \\ 0 & 0 & 0 \\ -\ddot{\theta}_1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ -l_2 \end{pmatrix}}_{\begin{pmatrix} -l_2 \ddot{\theta}_1 \\ 0 \\ 0 \end{pmatrix}} + \underbrace{S(\omega_2^2) \begin{pmatrix} 0 & 0 & \dot{\theta}_1 \\ 0 & 0 & 0 \\ -\dot{\theta}_1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ -l_2 \end{pmatrix}}_{\begin{pmatrix} -l_2 \dot{\theta}_1 \\ 0 \\ l_2 \dot{\theta}_1^2 \end{pmatrix}} \end{aligned}$$

$$\ddot{p}_{c_i}^i = \ddot{p}_i^i + S(\dot{w}_i^i) r_{i,c_i}^i + S(\dot{w}_i^i) S(\dot{w}_i^i) r_{i,c_i}^i$$

$$\ddot{p}_{c_2}^2 = \begin{pmatrix} l_1 | s_1 + 2 \dot{d}_2 \dot{\theta}_1 + d_2 \ddot{\theta}_1 - l_2 \ddot{\theta}_1 \\ 0 \\ -l_1 | c_1 + \ddot{d}_2 - d_2 \dot{\theta}_1^2 + l_2 \dot{\theta}_1^2 \end{pmatrix} = \begin{pmatrix} l_1 | s_1 + 2 \dot{d}_2 \dot{\theta}_1 + (d_2 - l_2) \ddot{\theta}_1 \\ 0 \\ -l_1 | c_1 + \ddot{d}_2 - (d_2 - l_2) \dot{\theta}_1^2 \end{pmatrix}$$

(c) Backward Recursion (Dynamics)

$$\tau_i = \begin{cases} f_i^{i^T} R_{i-1}^i \hat{K} & , q_i \text{ is prismatic} \\ \mu_i^{i^T} R_{i-1}^i \hat{K} & , q_i \text{ is revolute} \end{cases}$$

$$\dot{i}=2, \quad \tau_2 = f_2^{2^T} \underbrace{R_1^2}_{\mathbb{I}} \hat{K} = f_2^{2^T} \hat{K} = \hat{K}^T f_2^2$$

$$f_i^i = m_i \ddot{p}_{c_i}^i + R_{i+1}^i f_{i+1}^{i+1}$$

$$f_2^2 = m_2 \ddot{p}_{c_2}^2 + R_3^2 \overset{0}{f_3^3} = m_2 \begin{pmatrix} l_1 | s_1 + 2 \dot{d}_2 \dot{\theta}_1 + (d_2 - l_2) \ddot{\theta}_1 \\ 0 \\ -l_1 | c_1 + \ddot{d}_2 - (d_2 - l_2) \dot{\theta}_1^2 \end{pmatrix}$$

$$\tau_2 = -m_2 l_1 | c_1 + m_2 \ddot{d}_2 - m_2 (d_2 - l_2) \dot{\theta}_1^2 \quad \triangleleft$$

$$\dot{i}=1, \quad \tau_1 = \mu_1^{1^T} R_0^1 \hat{K} = \mu_1^{1^T} z_0^1 = \mu_1^{1^T} \hat{f} = \hat{f}^T \mu_1^1$$

$$R_0^1 = \begin{pmatrix} c_1 & s_1 & 0 \\ 0 & 0 & 1 \\ s_1 & -c_1 & 0 \end{pmatrix}$$

$$\mu_i^i = \mathbb{I}_i^i \dot{w}_i^i + S(\dot{w}_i^i) \mathbb{I}_i^i \dot{w}_i^i + R_{i+1}^i \mu_{i+1}^{i+1} + S(r_{i-1,c_i}^i) f_i^i - S(r_{i,c_i}^i) R_{i+1}^i f_{i+1}^{i+1}$$

$$\tau_1 = \hat{f}^T \mu_1^1 = \hat{f}^T \left[ \underbrace{\mathbb{I}_1^1 \dot{w}_1^1 + S(\dot{w}_1^1) \mathbb{I}_1^1 \dot{w}_1^1}_{H_1^1} + \underbrace{R_2^1}_{\mathbb{I}} \mu_2^2 + S(r_{0,c_1}^1) f_1^1 - S(r_{1,c_1}^1) \underbrace{R_2^1}_{\mathbb{I}} f_2^2 \right]$$

$$\hat{f} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$= \hat{f}^T \left[ \begin{pmatrix} * & * & * \\ * & \mathbb{I}_y^1 & * \\ * & * & * \end{pmatrix} \begin{pmatrix} 0 \\ \ddot{\theta}_1 \\ 0 \end{pmatrix} + \begin{pmatrix} * & * & * \\ 0 & 0 & 0 \\ * & * & * \end{pmatrix} H_1^1 + \begin{pmatrix} * \\ \mu_{2,y}^2 \\ * \end{pmatrix} + \begin{pmatrix} * & * & * \\ l_1 & 0 & 0 \\ * & * & * \end{pmatrix} \begin{pmatrix} f_{1,n}^1 \\ * \\ * \end{pmatrix} - \begin{pmatrix} * & * & * \\ l_1 & 0 & 0 \\ * & * & * \end{pmatrix} \begin{pmatrix} f_{2,n}^2 \\ * \\ * \end{pmatrix} \right]$$

$$r_{1,c_2}^2 = \begin{pmatrix} 0 \\ 0 \\ d_2 - l_2 \end{pmatrix}$$

$$r_{0,c_1}^1 = \begin{pmatrix} 0 \\ 0 \\ l_1 \end{pmatrix}$$

$$r_{1,c_1}^1 = \begin{pmatrix} 0 \\ 0 \\ l_1 \end{pmatrix}$$

$$\tau_1 = \mathbb{I}_{1,y}^1 \ddot{\theta}_1 + \mu_{2,y}^2 + l_1 (f_{1,n}^1 - f_{2,n}^2)$$

$$\mu_i^i = \mathbb{I}_i^i \dot{w}_i^i + S(\dot{w}_i^i) \underbrace{\mathbb{I}_i^i \dot{w}_i^i}_{H_i^i} + R_{i+1}^i \mu_{i+1}^{i+1} + S(r_{i-1,c_i}^i) f_i^i - S(r_{i,c_i}^i) R_{i+1}^i f_{i+1}^{i+1}$$

$$\mu_{2,y}^2 = \hat{f}^T \mu_2^2 = \left[ \begin{pmatrix} * & * & * \\ * & \mathbb{I}_y^2 & * \\ * & * & * \end{pmatrix} \begin{pmatrix} 0 \\ \ddot{\theta}_1 \\ 0 \end{pmatrix} + \begin{pmatrix} * & * & * \\ 0 & 0 & 0 \\ * & * & * \end{pmatrix} H_2^2 + \begin{pmatrix} * & * & * \\ d_2 - l_2 & 0 & 0 \\ * & * & * \end{pmatrix} \begin{pmatrix} f_{2,n}^2 \\ * \\ * \end{pmatrix} \right]$$

$$= \mathbb{I}_{2,y}^2 \ddot{\theta}_1 + (d_2 - l_2) f_{2,n}^2$$

$$f_{1,n}^1 = \hat{\lambda}^T f_1^1 = \hat{\lambda}^T \left[ m_1 \ddot{p}_{c_1}^1 + \underbrace{K_2^1}_{\mathbb{I}} f_2^2 \right] = \hat{\lambda}^T \left[ \begin{pmatrix} m_1 \ddot{p}_{c_{1,n}}^1 \\ \ddagger \\ \ddagger \end{pmatrix} + \begin{pmatrix} f_{2,n}^2 \\ \ddagger \\ \ddagger \end{pmatrix} \right]$$

$$= m_1 \ddot{p}_{c_{1,n}}^1 + f_{2,n}^2$$

$$\tau_1 = \bar{\mathbb{I}}_{1,y}^1 \ddot{\theta}_1 + \mu_{2,y}^2 + \ell_1 (f_{1,n}^1 - f_{2,n}^2)$$

$$\mu_{2,y}^2 = \bar{\mathbb{I}}_{2,y}^2 \ddot{\theta}_1 + (d_2 - \ell_2) f_{2,n}^2$$

$$\tau_1 = \bar{\mathbb{I}}_{1,y}^1 \ddot{\theta}_1 + \bar{\mathbb{I}}_{2,y}^2 \ddot{\theta}_1 + (d_2 - \ell_2) f_{2,n}^2 + \ell_1 (m_1 \ddot{p}_{c_{1,n}}^1 + \cancel{f_{2,n}^2} - \cancel{f_{2,n}^2})$$

$$\ddot{p}_{c_1}^1 = \begin{pmatrix} |g|s_1 + \ell_1 \ddot{\theta}_1 \\ 0 \\ -|g|c_1 - \ell_1 \dot{\theta}_1^2 \end{pmatrix}, \quad f_2^2 = m_2 \begin{pmatrix} |g|s_1 + 2d_2 \dot{\theta}_1 + (d_2 - \ell_2) \ddot{\theta}_1 \\ -|g|c_1 + \ddot{d}_2 - (d_2 - \ell_2) \dot{\theta}_1^2 \end{pmatrix}$$

$$\tau_1 = (\bar{\mathbb{I}}_{1,y}^1 + \bar{\mathbb{I}}_{2,y}^2) \ddot{\theta}_1 + (d_2 - \ell_2) m_2 [|g|s_1 + 2d_2 \dot{\theta}_1 + (d_2 - \ell_2) \ddot{\theta}_1] + \ell_1 m_1 [\ell_1 \ddot{\theta}_1 + |g|s_1]$$

$$\tau_1 = (\bar{\mathbb{I}}_{1,y}^1 + \bar{\mathbb{I}}_{2,y}^2 + m_2 (d_2 - \ell_2)^2 + m_1 \ell_1^2) \ddot{\theta}_1 + 2m_2 (d_2 - \ell_2) \dot{\theta}_1 \dot{d}_2 + [m_1 \ell_1 + m_2 (d_2 - \ell_2)] |g|s_1$$

$$\tau_2 = -m_2 |g|c_1 + m_2 \ddot{d}_2 - m_2 (d_2 - \ell_2) \dot{\theta}_1^2$$

$$\tau = \underbrace{\begin{pmatrix} \bar{\mathbb{I}}_{1,y}^1 + \bar{\mathbb{I}}_{2,y}^2 + m_1 \ell_1^2 + m_2 (d_2 - \ell_2)^2 & 0 \\ 0 & m_2 \end{pmatrix}}_B \underbrace{\begin{pmatrix} \ddot{\theta}_1 \\ \ddot{d}_2 \end{pmatrix}}_{\phi = C\dot{q}} + \underbrace{\begin{pmatrix} 2m_2 (d_2 - \ell_2) \dot{\theta}_1 \dot{d}_2 \\ -m_2 (d_2 - \ell_2) \dot{\theta}_1^2 \end{pmatrix}}_{\phi} + \underbrace{\begin{pmatrix} [m_1 \ell_1 + m_2 (d_2 - \ell_2)] |g|s_1 \\ -m_2 |g|c_1 \end{pmatrix}}_g$$

$$\tau = \tau(q, \dot{q}, \ddot{q})$$

$$B(q) = \frac{\partial \tau}{\partial \ddot{q}} = \begin{pmatrix} \frac{\partial \tau_1}{\partial \ddot{\theta}_1} & \frac{\partial \tau_1}{\partial \ddot{d}_2} \\ \frac{\partial \tau_2}{\partial \ddot{\theta}_1} & \frac{\partial \tau_2}{\partial \ddot{d}_2} \end{pmatrix}, \quad g(q) = \tau(q, 0, 0)$$

$$\phi(q, \dot{q}) = C(q, \dot{q}) \dot{q} = \tau(q, \dot{q}, 0) - g(q)$$